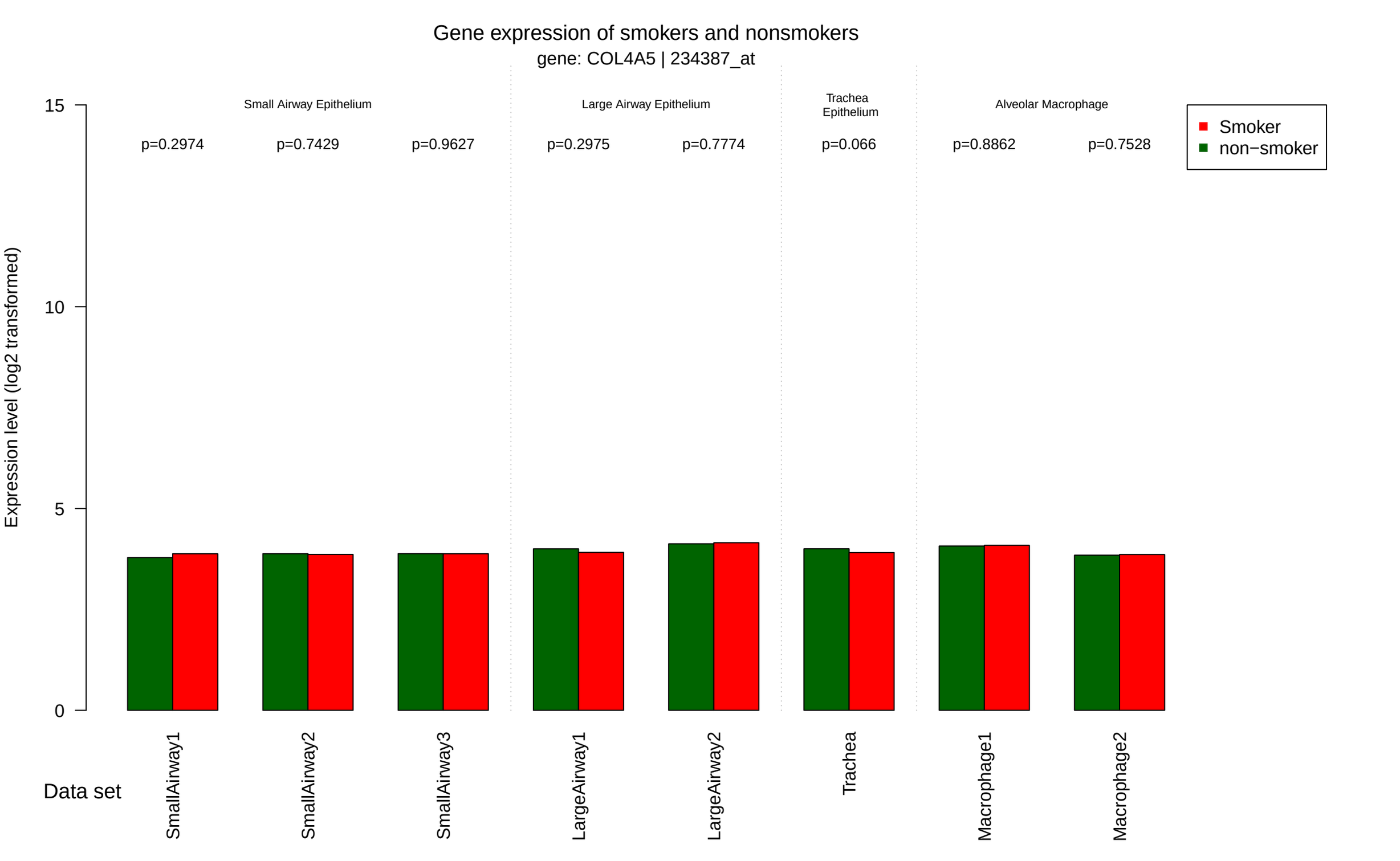
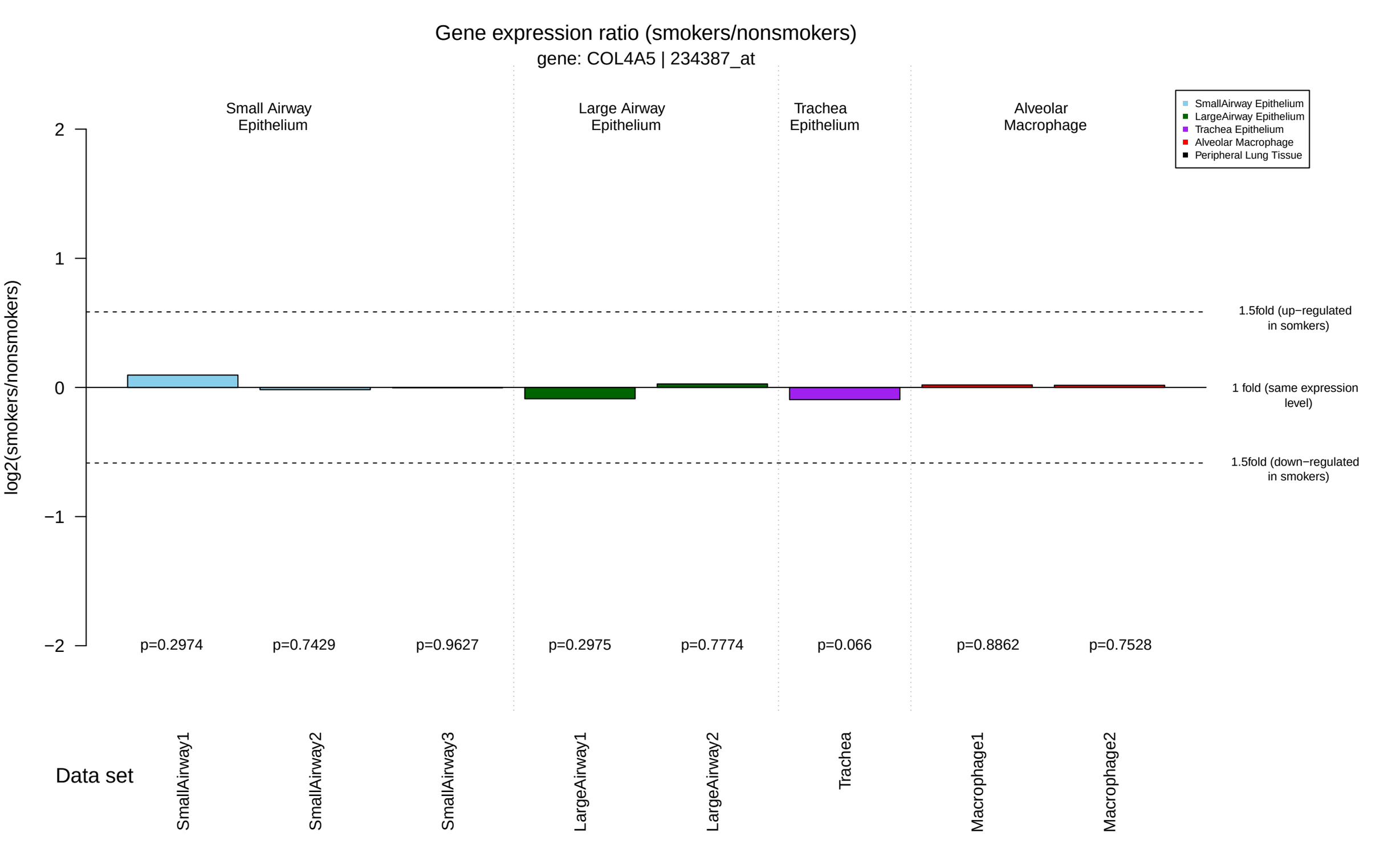
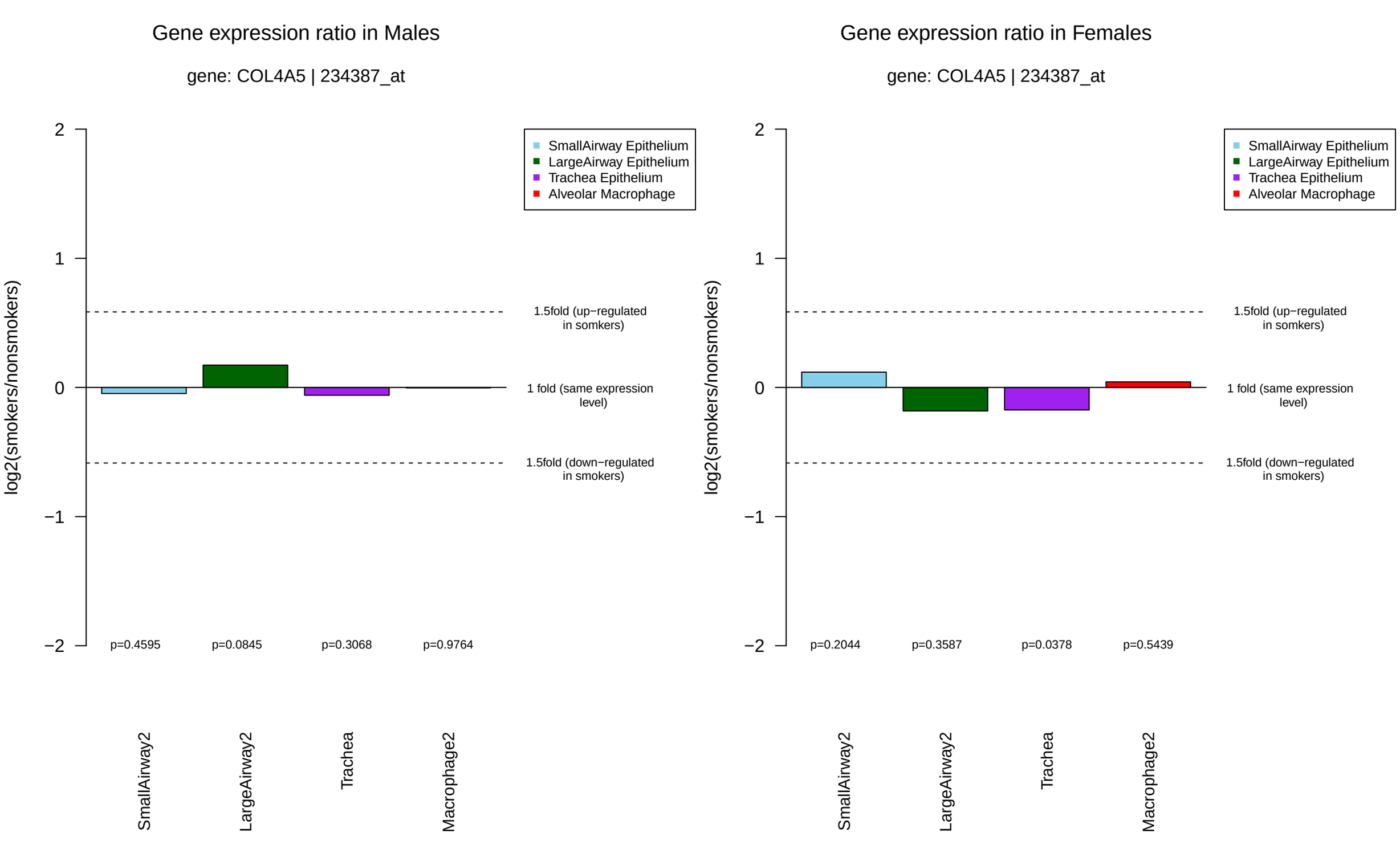
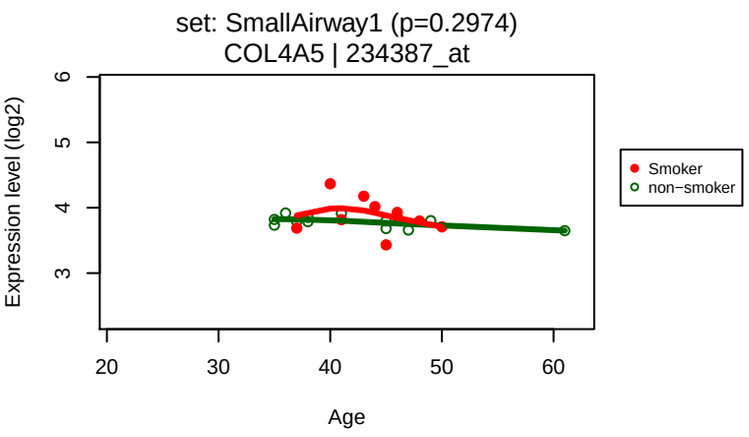
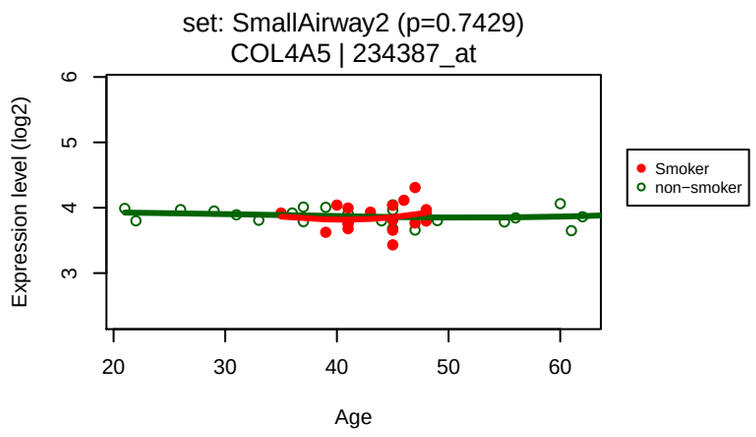
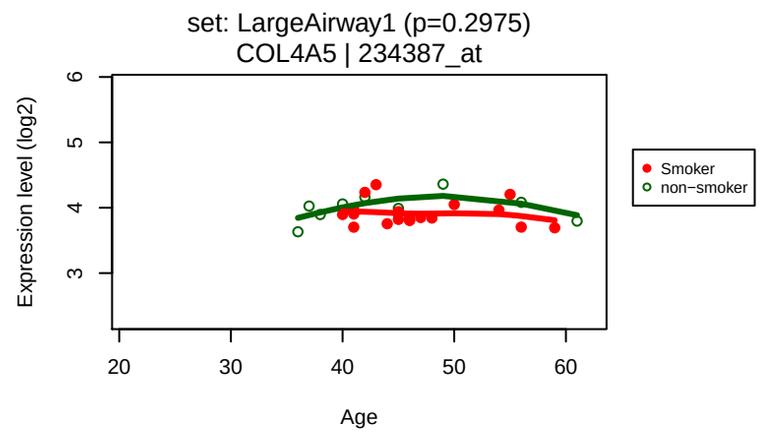

Gene expression of smokers and nonsmokers
gene: COL4A5 | 234387_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2974
p=0.7429
p=0.9627
p=0.2975
p=0.7774
p=0.066
p=0.8862
p=0.7528
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

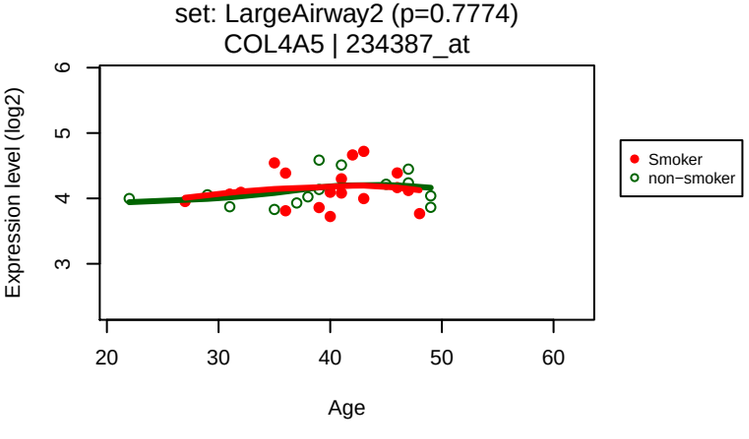

Gene expression ratio (smokers/nonsmokers)
gene: COL4A5 | 234387_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2974
p=0.7429
p=0.9627
p=0.2975
p=0.7774
p=0.066
p=0.8862
p=0.7528
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

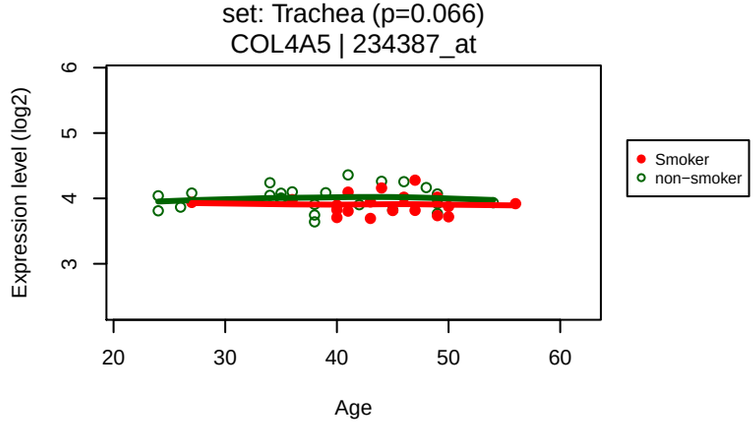

Gene expression ratio in Males
gene: COL4A5 | 234387_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4595
p=0.0845
p=0.3068
p=0.9764
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A5 | 234387_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2044
p=0.3587
p=0.0378
p=0.5439
SmallAirway2
LargeAirway2
Trachea
Macrophage2

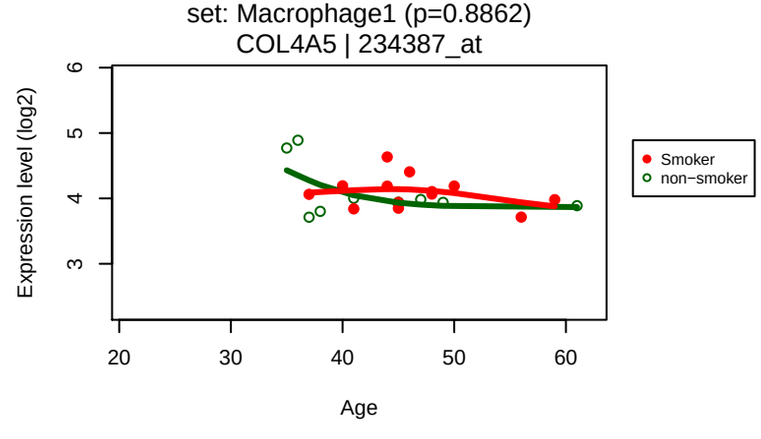

set: SmallAirway1 (p=0.2974)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker

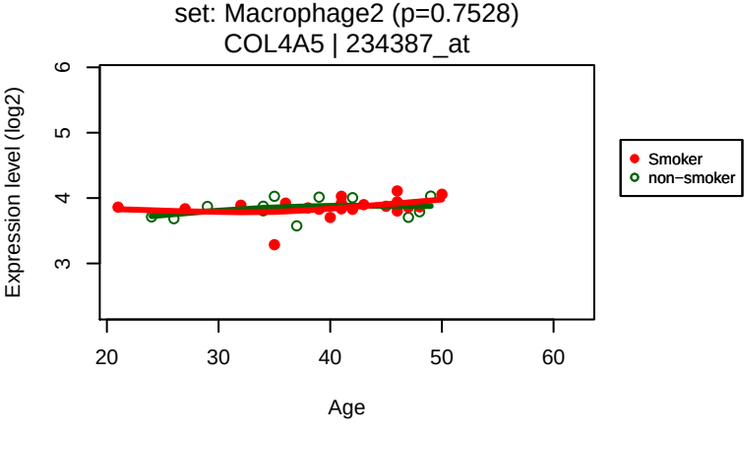

set: SmallAirway2 (p=0.7429)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker

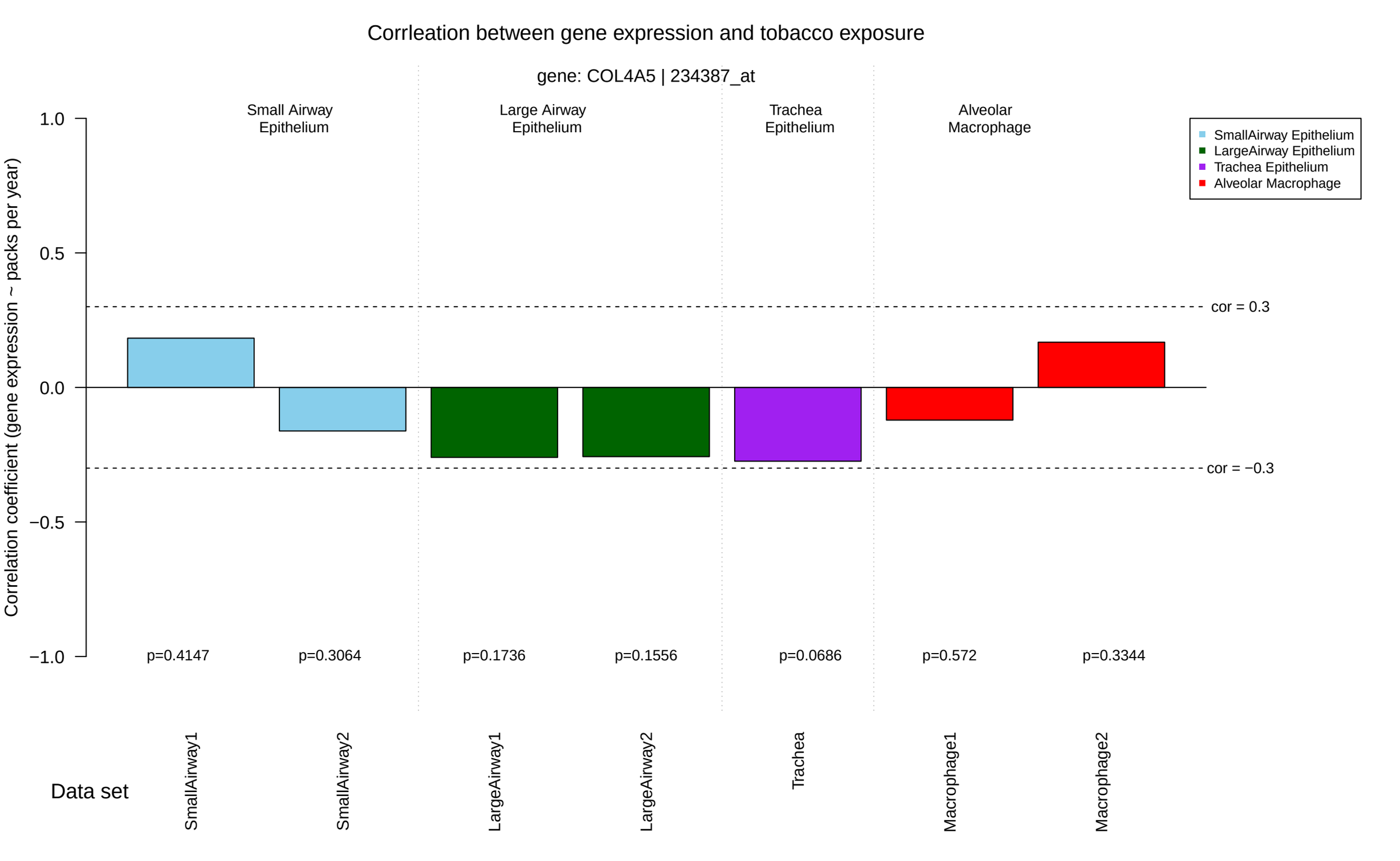

set: LargeAirway1 (p=0.2975)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7774)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.066)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8862)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7528)
COL4A5 | 234387_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A5 | 234387_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4147
p=0.3064
p=0.1736
p=0.1556
p=0.0686
p=0.572
p=0.3344
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2